

Motor Imagery Classification in EEG signals using Hybrid CNN-ISqueeze net

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Abstract— Motor imagery (MI) electroencephalography (MI-EEG) data are widely used in BCI systems to determine participant intention. MI classification is challenging because of several reasons, such as poor signal-to-noise ratios (SNR) and lack of excellent data. The operation of BCI devices requires an understanding of MI-EEG signals. Recent years have seen the effective use of deep learning (DL) techniques in recognizing patterns and other domains. Conversely, there aren't many efficient DL algorithms that have been used with BCI systems. This research work presents a novel MIC scheme using the hybrid classifier. At first, the EEG signals are pre-processed by applying the band pass filtering (BPF) method. Then, features including Stockwell transform (ST), correntropy and Common Spatial Pattern (CSP) features are extracted. For MIC, a hybrid classification method that combines CNN and Improved squeeze net (ISQN) is employed in this work.

Keywords—Brain-Computer Interface; EEG signal; Motor Imagery; Correntropy; Improved Squeeze net.

Nomenclature

Abbreviation	Description
ANFIS	Adaptive Neuro-Fuzzy Inference System
BPF	Band Pass Filtering
BCI	Brain-Computer Interface
CSP	Common Spatial Pattern
CNN	Convolutional Neural Networks
CWT	Continuous Wavelet Transform
DEC	Deep EEG-Channel-Attention
DL	Deep Learning
DCNN	Deep Convolutional Neural Networks
EEG	Electroencephalography
EMD	Empirical Mode Decomposition
FC	Fully Connected
HD	High-Dimensional
IMF	Intrinsic Mode Functions
ISQN	Improved Squeeze Net
iTFE	Initial Temporal Feature Extraction
JFOFL	Jellyfish Optimization with Fuzzy Logic
LSTM	Long Short-Term Memory Term
MI	Motor Imagery
MCCSP	Multi-Class Common Spatial Patterns
MIC	Motor Imagery Classification
SVM	Support Vector Machine
ST	Stockwell Transform
SQN	Squeeze Net
SNR	Signal-to-Noise Ratio
3D-CapsNet	Three-Dimensional Capsule Network
WTS-CC	Wavelet-Based Temporal-Spectral-Attention Correlation Coefficient

I. INTRODUCTION

With the technology known as brain-computer interface, or BCI, an intermediary is not necessary while communicating between the human brain and exterior equipment. It enables direct brain-generated control of a PC or smart device, avoiding the need for muscles and other peripheral organs [1] [2] [3]. BCI research is widely applied in various fields. Research on BCI is very valuable in many different domains. BCI provides extraordinary insight into

operational human minds. In the field of medicine, it is utilized to cure ailments like paralysis that were previously thought to be incurable [4] [5] [6]. The EEG signal is a widely used brain waves in BCI as it is portable, inexpensive, non-invasive and has a higher temporal resolution. Having many EEG channels may produce additional details regarding the cerebral activity in the background, which also increases noise-induced redundancy.

Along with the previously stated factors, the practicality of implementing BCI like the decrease in BCI costs, and the enhancement of BCI performance are the key factors of reducing the amount of EEG signal channels. Numerous analyses have suggested channel selection techniques for MI-based BCI for addressing the issue of performance degradation by noise and replicated information across various EEG channels. Different approaches have been implemented for MI to choose the proper EEG signal channel [7] [8] [9]. Wrapper-oriented and filter-oriented approaches are the two categories into which channel selection techniques fall. Channels are frequently elected by means of filter methods based on F score, or other criteria [10] [11].

The effectiveness of several individual and combined classifiers has already been assessed with regard to the categorization of MI EEG signals [12] [13]. Previous research evaluated the effectiveness of individual and combined classifiers currently in use of real-time BCI applications and other related applications [28][29] [30]. It is commonly recognized that the large dimension and dynamic characteristics of actual-time EEG data make the present classifiers less accurate. The poor conductance of the electrodes on the scalp might occasionally cause the signals to be skewed with noise and artifacts [14] [15] [16]. Furthermore, the poor SNR and excessively weak high-dimensional EEG data present challenges. Thus, it is needed to develop a new MIC model, and accordingly, this work presents a hybrid model for MI classification. The contributions are as follows:

- Develops a new MIC model, where, ST, CSP and correntropy features are extracted.
- Introduces a hybrid model that combines the CNN with the ISQN model for the classification of MI in EEG signals.

The review on MIC in EEG is given under section II. A short explanation of MIC in EEG is in section III. Pre-processing and feature extraction are under section IV. CNN and SQN are in Section V. Section VI depicts the results

II. LITERATURE REVIEW

A. Related works

In 2023, Hsu and Cheng [17] presented a text MI EEG discriminating approach termed as WTS-CC that considered temporal, spectral, spatial, and EEG channel domain concurrently. The important temporal aspects were derived using the iTFE module. Then, it was suggested that the

DEC module automatically modified the weight of every channel and efficiently amplified the EEG channel. Subsequently, the WTS module was presented to get more significant discriminatory features between different MI tasks. Lastly, MI EEG discrimination was accomplished using a basic discrimination module. From the results, the suggested text WTS-CC approach surpassed extant techniques on accurate classification and produce promising discriminating performance.

A 3D-CapsNet identification framework was presented in 2023 by Du *et al.* [18] for extracting the S&T features found in EEG signals to facilitate the recognition of MI. This model integrated the S&T dimensions, and the inherent connection among features. Initially, low-level features were extracted using a multi-layer 3D convolution module. Second, the integration of capsule networks also yields improved spatial properties. Lastly, squash functions and dynamic routing links were used for categorization. All of the subjects' datasets showed good performance using 3D-CapsNet model.

In 2023, Alizadeh *et al.* [19] provided a novel method for categorizing multi-class MI-EEG data. It consisted of MCCSP and an EMD based signal processing algorithm. In particular, the optimal IMF were suggested to replace the EEG signal for subsequent processing after applying the EMD. The CSP method was then extended to the MCCSP, which was used as the tool for extracting features. The results showed that when MCCSP was deployed precisely to EEG channel data, the suggested approach greatly increased classification accuracy.

In 2022, Jing *et al.* [20] offered a new MI classification model depending upon graph theory and the evaluation of neurological connectivity among various brain areas. In particular, themes that depict the brain's local network architecture were taken out. Next, a classifier for identifying essential structures in the graph was constructed using an Ego-CNNs graph embedding model. The proposed method demonstrated that the method generated high classification accuracies in both multiclass (90.33% for dataset 1) and 2-class tasks. Compared to previous ways, the suggested method got an average Kappa score of 0.88 over nine individuals, which was better.

In 2024, Eunmok *et al.* [21] introduced the JFOFL method for MIC BCI that attempted to categorize EEG motor imaging signals by utilizing fuzzy logic with metaheuristics. First, it performed the CWT to convert 1D-EEG data into 2D time-frequency amplitude signals. The JFOFL-MICBCI methodology employed the SQN method for feature extraction, and the JFO system for modifying the hyperparameters. The ANFIS technique was utilized to carry out the classification procedure. The improved efficiency of the JFOFL-MICBCI approach was demonstrated using a wide range of trials.

III. MI CLASSIFICATION IN EEG SIGNALS: A SHORT EXPLANATION

MI is the term for a mental activity in which people visualize moving certain bodily parts without actually moving them. MI has been intensively researched to determine patients' intentions for movement who have restricted possibilities for movements. The proposed MIC model in EEG Signals involves the following processes:

Pre-processing- The raw EEG signal is pre-processed by applying the BPF method.

Feature Extraction- Here, we extract the features like Stockwell transform, correntropy and CSP.

Classification- A hybrid classifier that combines models like CNN and ISQN is proposed to do the classification.

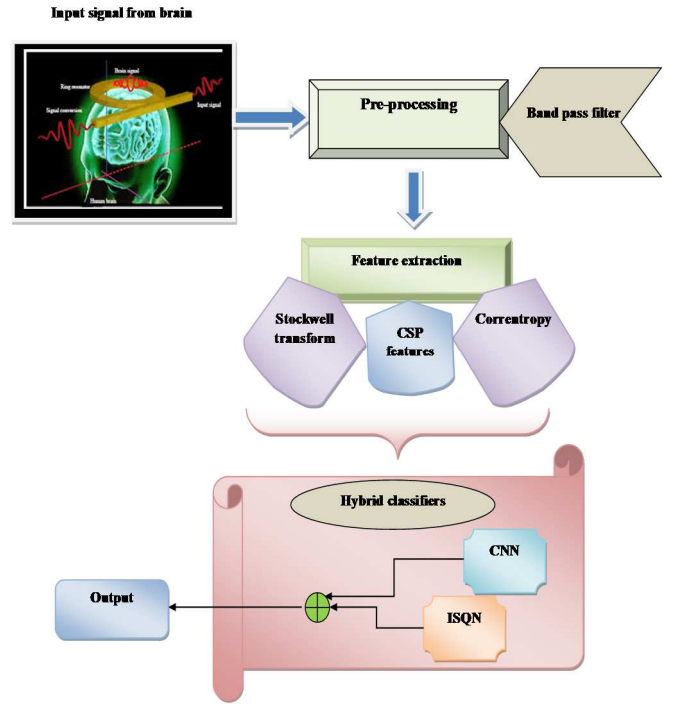


Fig. 1. Architecture of proposed MIC model in EEG Signals

IV. PRE-PROCESSING AND FEATURE EXTRACTION

A. Pre-processing

The EEG signal pre-processing involves the conversion of raw EEG data into clean ones by removing unsolicited noise that is suited for subsequent processing.

Let the raw signal be S , which is pre-processed using the BPF approach.

BPF Technique: Bandpass filtering [22] is a fundamental preprocessing step in EEG (electroencephalogram) signal analysis. EEG signals typically cover a broad frequency spectrum (from a few Hz to several tens of Hz), and different frequency bands often contain distinct physiological information. The signal that is preprocessed is denoted as S^{BPF} .

B. Feature Extraction

The process of attaining the pertinent data from signals is known as feature extraction. In this work, we extract the subsequent features:

- ♣ Stockwell transform
- ♣ Correntropy and
- ♣ CSP features

Stockwell transform: This [23] is a time-frequency analysis technique that provides both time and frequency localization of signals S^{BPF} , which is modeled as in Eq. (1). ST captivates the convolution of $S^{BPF}(t)$ using a window function $g(t-\tau)$, allowing accurate localization of frequency component.

$$ST(\tau, f) = \int_{-\infty}^{+\infty} S^{BPF}(t) g(t-\tau) \exp(-i2\pi ft) dt \quad (1)$$

Eq. (2) shows the formula for ST with a Gaussian window function.

$$ST^*(\delta, \tau, f) = \int_{-\infty}^{+\infty} S^{BPF}(t) \frac{1}{\sqrt{2\pi}\delta} \exp\left(-\frac{(\tau-t)^2}{2\delta^2}\right) \exp(-i2\pi ft) dt \quad (2)$$

The ST ensures thorough coverage of both high and low-frequency elements in S^{BPF} by dynamically adjusting the window width dependent on signal frequency. Equation (3) expresses the formula for adaptive window width.

$$\delta(f) = \frac{1}{|f|} \quad (3)$$

Finally, the extracted ST features are denoted as fe^{ST} .

Advantages of Stock Well Features

- Non-stationary signal analysis is made possible by the Stockwell Transform, which offers a joint time-frequency representation.
- This is especially important for signals like EEG that fluctuate over time.
- It is useful for spotting fleeting events in signals because it provides strong temporal and frequency precision.

Correntropy [24]: It calculates the level of similarity amid 2 arbitrary variables and turns every sample into uncorrelated data to increase its resistance to noise and outliers. Eq. (4) represents the mathematical definition of correntropy. In this case, n denotes a collection of samples mentioned by $(U_i, V_i)_{i=1}^N$ and U, V are the two arbitrary variables.

$$fe^{CR} = Corr(U, V) = \frac{1}{n} \sum_{i=1}^n H_{\sigma}(U_i - V_i) \quad (4)$$

The Gaussian kernel $H_{\sigma}(U_i - V_i)$ is defined as in Eq. (5), where, σ is employed for maintaining the width of kernel constraints.

$$H_{\sigma}(U_i - V_i) = \exp\left(-\frac{\|U_i - V_i\|^2}{2\sigma^2}\right) \quad (5)$$

Advantages of Correntropy Features

- Nonlinear dependencies between signals are captured by entropy, which is useful in complicated signal situations when linear approaches are ineffective.
- It can extract features from distorted signals with greater accuracy since it is less susceptible to noise and outliers.
- The ability to compute entropy at several scales facilitates the capture of characteristics at varying resolutions.

CSP: It [25] is a potent method to extract features in EEG signals that is utilized in BCI devices. As it helps to improve the discriminatory data contained in EEG signals, it is very helpful for brain activity detection tasks. CSP makes it simpler to discern amid various mental activities or states. Here, the input is S^{BPF} . Consider the following two types of preprocessed signals $S_{i,1}^{BPF}, S_{i,2}^{BPF} \in Z^{N \times T}$ from the experiments, where, T denotes the sample points and N denotes number of channels. For a class cl , the matrix for spatial covariance is found using Eq. (6).

$$D_m = \frac{1}{n_{cl}} \sum_{i=1}^{n_{cl}} S_{i,cl}^{BPF} S_{i,cl}^{BPF^T} \quad (6)$$

The total amount of trails in cl is indicated as n_{cl} in Eq. (6). Eq. (7) may be used to calculate how the spatial filter would affect one class's variance while decreasing the other variance.

$$K_{cl}(\vartheta) = \frac{\vartheta^T D_m \vartheta}{\vartheta^T D_m \vartheta} \quad \text{where, } \|\vartheta\|_2 = 1 \quad (7)$$

The spatial filter from Eq. (7) is represented by ϑ . Subsequently, the Rayleigh quotient maximization may be converted into a solution for the general eigenvalue issue using Eq. (8).

$$D_m \vartheta = \chi D_m \vartheta \quad (8)$$

In Eq. (8), the eigenvector and eigenvalue are denoted as ϑ and χ . The eigenvectors related to m greatest and lowest eigenvalues are used to build the spatial filters Sp_{CSP} . Eq. (9) therefore expresses the resultant projection signal Y for a single trial.

$$Y = Sp_{CSP}^T X \quad (9)$$

Eq. (10) shows the specific trial for p th feature.

$$fe^{CSP} = \log\left(Y_{p^{feature}} Y_{p^{feature}}^T\right) \quad (10)$$

The total features derived for MIC in EEG signals are denoted by $fe = [fe^{CSP} fe^{ST} fe^{CR}]$.

Advantages of CSP Features

- It maximizes the variance of various classes while minimizing the amount of characteristics, which is very helpful for categorizing signals.
- Understanding the spatial patterns linked to various mental states can be facilitated by CSP, which also helps to clarify the underlying neuronal processes.
- It is a dependable option because it typically performs effectively under a variety of circumstances and situations.

V. HYBRID CLASSIFICATION MODEL: COMBINING CNN AND IMPROVED SQUEEZENET FOR MIC

The derived features fe is subjected to a hybrid classification model combining CNN and ISQN for MIC that are explained in the below sections. The outcomes from both models are combined to determine the final classification results.

A. CNN

The CNN architecture [26] includes varied layers: "Convolutional layer, pooling layer, flatten layer, fully connected layer and dropout layer". Convolutional Layers extract different information from the input fe by performing convolutions by means of learnable filters. To add non-linearities, activation functions such as ReLU are employed after convolutions. Pooling layers preserve significant information while reducing the spatial aspects of the convolved features. The two most popular pooling techniques for down-sampling features are average pooling and max pooling.

With the help of the features from convolution and pooling layers, the FC layer carries out high-level reasoning. This layer is typically linked to the output layer in MIC models. Convolutional and pooling layer output is

transformed into a 1D vector by flattening layers, which are then fed into dense layers. To avoid overfitting, dropout layers arbitrarily neutralize a portion of neurons throughout training. Consequently, the CNN result is demonstrated as $O(CNN)$.

B. Improved Squeezenet

The SQN [27] architecture highly aids in MCI in EEG signals. In SQN design, 'conv1' and 'conv10' are the first and final convolution layers respectively. Eight Fire components make up SQN, which improves feature extraction abilities. SQN also includes max-pooling functions with a stride of 2 after "conv1, fire 4, fire 8, and conv10" to help in feature selection and dimensionality reduction. Models like Squeezenet suffer from overfitting, especially in complex datasets like those used for detection/classification in EEG signals. Implementing a 50% dropout ratio after the 'fire 9' module helps in regularizing the model. Dropout randomly deactivates neurons during training, which prevents the model from relying too heavily on certain pathways and thus improves generalization to new data. The existing SQN model is shown in Fig. 2.

Even though SQN is well-suited for detecting tasks, it poses certain drawbacks, while considering pooling operation. In traditional SQN, we have max pooling layer, which is not appropriate for every application and data kind. This issue is sorted out by our proposed ISQN model, where, we have deployed a custom mixed pooling layer instead of max pooling. The developed SQN model includes 1 input layer, 1 BN layer, 9 fire modules and 8 custom mixed pooling layers. The input layer gets the feature set fe as input. Next comes the batch normalization. The BN permits utilization much advanced learning rates that further rises the speed of the training network. After BN, a fire module is used. After each fire module, a custom mixed pooling layer is employed.

Traditional Squeezenet uses max pooling, which is not suitable for all applications and data types. Therefore, this research work introduces a custom mixed pooling layer instead of max pooling to help in better handling activations. This layer combines max pooling with average pooling, which down-weights non-maximal activations. This adaptation allows for a more nuanced feature selection and potentially better representation of complex EEG signal patterns.

The proposed custom mixed pooling layer is modelled as in Eq. (11), where, A_i denotes output from the conv layer and r_j denotes maximal feature size.

$$PCMP = \chi \cdot \max_{i \in r_j} A_i + (1 - \chi) \cdot \frac{1}{|r_j|} \sum_{i \in r} A_i \quad (11)$$

The scaling factor χ is evaluated as in Eq. (12), where, f_j denotes fire module output and R_j denotes a random integer between 0 and 1.

$$\chi = 1 + e^{-\sum_{i=1}^M f_j \cdot R_j} \quad (12)$$

The proposed custom mixed pooling layer could learn adaptively from the input and improve the classification performance. The ISQN output is denoted by $O(SQN)$.

The outputs $O(CNN)$ and $O(SQN)$ model are averaged that provides the result on MIC in EEG signals.

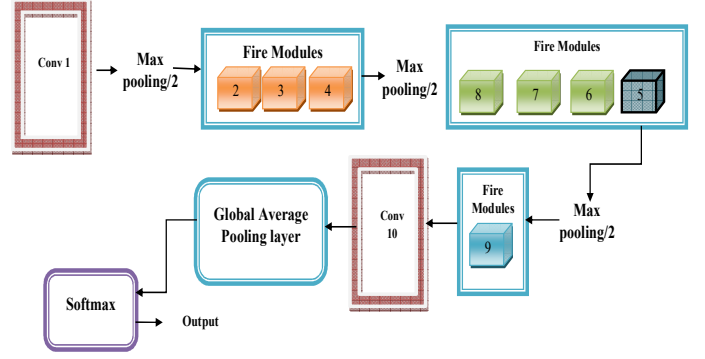


Fig. 2. Conventional SQN Architecture

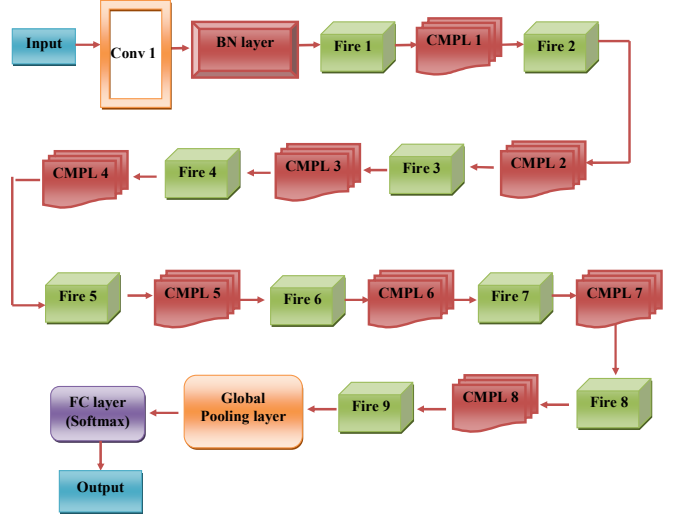


Fig. 3. Improved Squeeze net Architecture

VI. RESULTS AND DISCUSSION

A. Experimental procedure

The proposed MIC model in EEG signals was implemented in Python using the Hybrid CNN-ISQN model. "In addition, we simulate a system equipped with a 64-bit OS, x64 based processor and had 16.0 GB (13.9 GB usable) of RAM". The scrutiny on improved Squeeze net was measured over SVM, LSTM, Bi-LSTM, LENet, DCNN, CNN and SQN models. The dataset was collected from [31].

Dataset description: "The EEG signal with the following subject is used for evaluation: "A01T, A01E, A02T, A02E, A03T, A03E, A04T, A04E, A05T, A05E, A06T, A06E, A07T, A07E, A08T, A08E, A09T, A09E". Moreover, the subjects "A01E, A02E, A03E, A04E, A05E, A06E, A07E, and A08E" are used for training; while, the rest of the subject is used for testing".

B. Performance analysis

Fig. 4, Fig. 5 and Fig 6 show the performance of Hybrid CNN-ISQN over prevailing models like LSTM, LENet, SVM, Bi-LSTM, SQN, CNN and DCNN. For better MIC in EEG signals, the Hybrid CNN-ISQN must reach better results over compared SVM, LSTM, Bi-LSTM, LENet, DCNN, CNN and SQN. The outcomes have shown that Hybrid CNN-ISQN bettered LSTM, LENet, SVM, Bi-LSTM, SQN, CNN and DCNN for all measured cases. The precision noticed from Fig. 4(c) shows that Hybrid CNN-ISQN attains higher value of 0.9 for A01T, while, LSTM, LENet, SVM, Bi-LSTM, SQN, CNN and DCNN attains less precision of 0.7816, 0.67816, 0.7126, 0.7356, 0.839, 0.7586 and 0.7931. In proposed custom mixed pooling layer, we

add average pooling that down weighs the activation by merging the non-maximal activation. This improves the precision of classification. The MCC using Hybrid CNN-ISQN at A01T is high around 0.87, while, LSTM, LENet,

SVM, Bi-LSTM, SQN, CNN and DCNN attains less MCC values of 0.70231, 0.564 0.6107, 0.64124, 0.7786, 0.6717 and 0.71757 respectively.

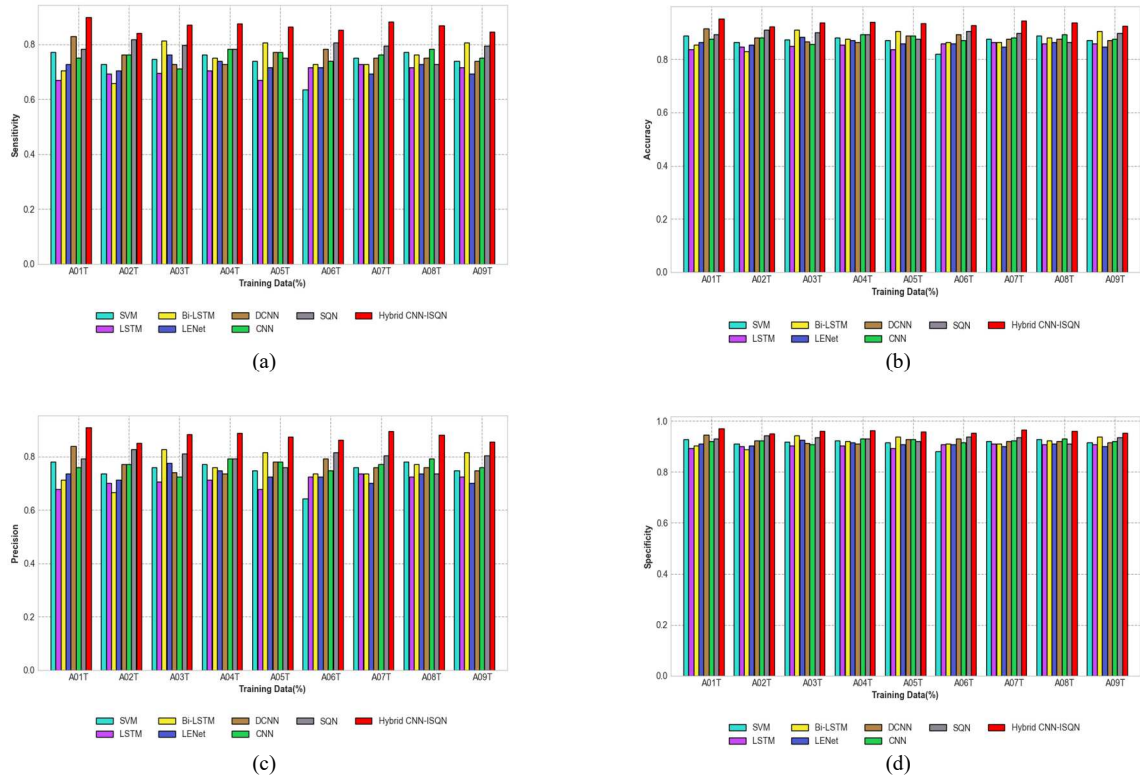


Fig. 4. Analysis of Hybrid CNN-ISQN method over prevailing approaches for MIC on (a) Sensitivity (b) Accuracy (c) Precision and (d) Specificity

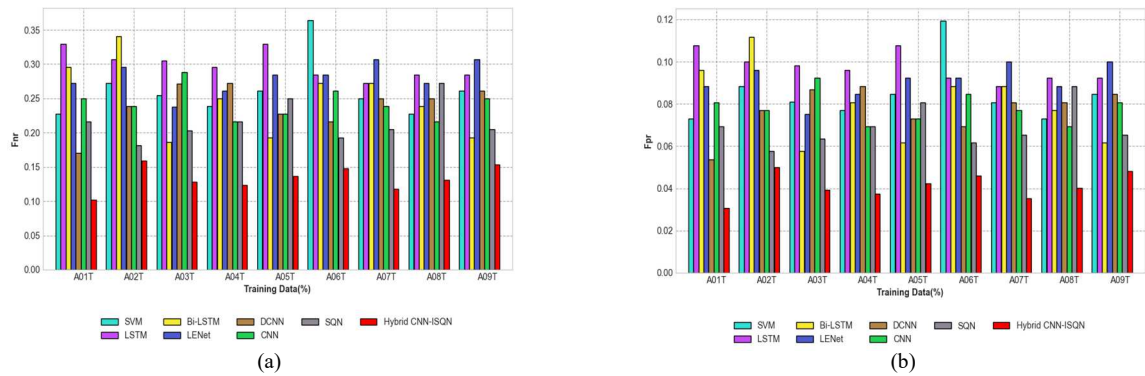
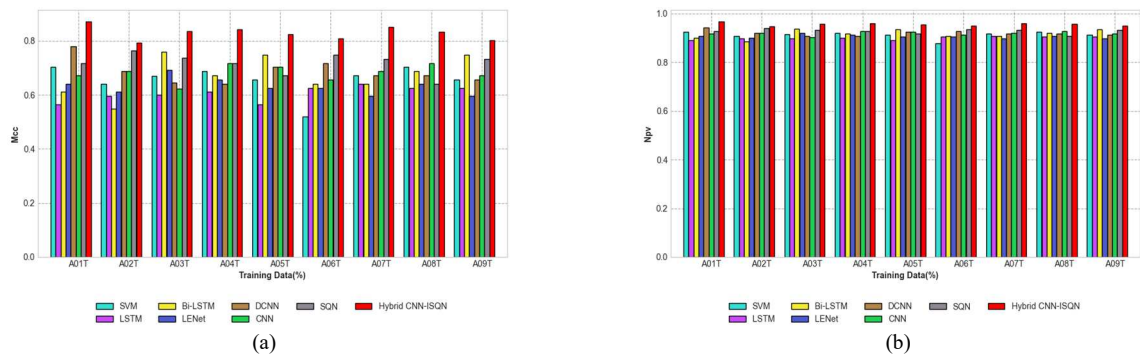


Fig. 5. Analysis on Hybrid CNN-ISQN method over prevailing approaches for MIC on (a) FNR (b) FPR



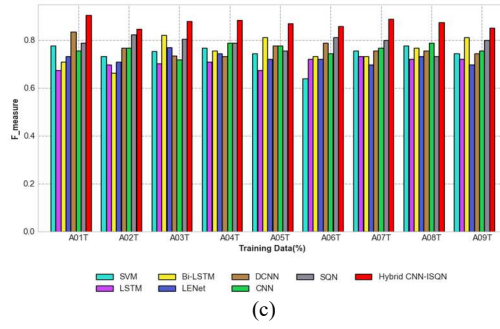


Fig. 6. Analysis of Hybrid CNN-ISQN method over prevailing approaches for MIC on (a) MCC (b) NPV and (c) F-measure

C. Statistical analysis

The statistical study is shown in Table I based on accuracy metric. The outcomes on all cases show that Hybrid CNN-ISQN attains high MIC accuracy over LSTM, LENet, SVM, Bi-LSTM, SQN, CNN and DCNN. The accuracy attained for the maximal case is around 0.951149, while, LSTM, LENet, SVM, Bi-LSTM, SQN, CNN and

DCNN attains relatively less accuracy of 0.8879, 0.864, 0.90, 0.8836, 0.9167, 0.894 and 0.91092 respectively. In addition, for the median case, Hybrid CNN-ISQN attains a high MIC accuracy of 0.936785. The proposed custom mixed pooling layer improves the abilities of SQN, thus enhancing the betterment of MIC in EEG signals.

TABLE I. STATISTICAL STUDY ON HYBRID CNN-ISQN METHOD OVER PREVAILING APPROACHES FOR MIC IN EEG SIGNALS

Metrics	SVM	LSTM	Bi-LSTM	LENet	DCNN	CNN	SQN	Hybrid CNN-ISQN
Max	0.887931	0.864943	0.909483	0.883621	0.916667	0.893678	0.91092	0.951149
SD	0.019682	0.009716	0.025124	0.010818	0.015266	0.010818	0.013621	0.008722
Median	0.875	0.853448	0.876437	0.859195	0.876437	0.882184	0.899425	0.936785
Min	0.818966	0.836207	0.83046	0.847701	0.864943	0.857759	0.864943	0.922414
Mean	0.87053	0.851692	0.876916	0.861271	0.881705	0.880109	0.893838	0.935629
variance	0.000387	9.44E-05	0.000631	0.000117	0.000233	0.000117	0.000186	7.61E-05
Skewness	1.819623	0.473412	0.257868	0.579676	1.085449	0.579676	0.978629	0.104754
Kurtosis	5.385909	1.940666	2.064303	2.671857	3.40955	2.671857	2.888309	2.083141

D. Ablation Study

Fig 7 shows the ablation study analysis. The accuracy of the proposed model is compared with the without feature extraction, proposed with stock well transform features, proposed with CSP features and proposed with correntropy features. The accuracy of the proposed model is 0.95. Hence the betterment of the proposed model is proved.

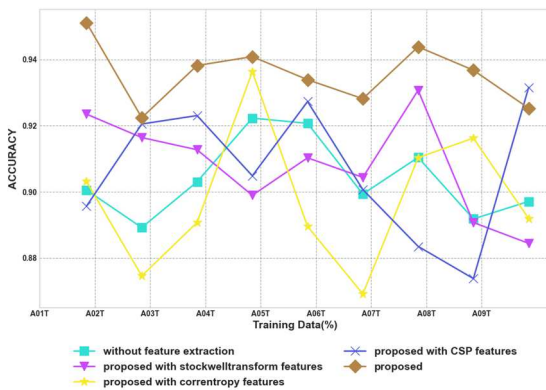


Fig. 7. Ablation Study Analysis

VII. CONCLUSION

This research presented a novel MIC scheme using hybrid classifiers. At first, the EEG signals were pre-processed by applying the BPF method. Then, features including ST, corr-entropy and CSP features were derived. For MIC, hybrid methods such as CNN and ISQN were employed that provided MIC output. From analysis, the Hybrid CNN-ISQN attained the higher precision value of 0.9 for A01T, while, LSTM, LENet, SVM, Bi-LSTM, SQN, CNN and DCNN attained less precision of 0.7816, 0.67816, 0.7126, 0.7356, 0.839, 0.7586 and 0.7931. The MCC using Hybrid CNN-ISQN at A01T was high around 0.87, while, LSTM, LENet, SVM, Bi-LSTM, SQN, CNN and DCNN attained less MCC values of 0.70231, 0.564, 0.6107, 0.6412, 0.7786, 0.67177 and 0.71757 respectively. The current focus of the research is on motor imagery classification using customized classifiers. However, for future directions, it will be crucial to explore more advanced deep learning models. Validating these models through ablation studies will be essential to demonstrate their effectiveness and superiority in motor imagery classification tasks.

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